מכון טכנולוגי חולון
Holon Institute of Technology

הפקולטה למדעים
מערך פרויקטים ושיתוף פעולה עם התעשייה

SHADE SYSTEM

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Table of Contents:

1) Introduction 3-4
   a. Background & Problem Definition 3
   b. Value Proposition and Justification 3
   c. Key Terms and Definitions 4
2) System Description 5-21
   a. Project Overview 5
   b. Feasibility Assessment 5
   c. Stakeholder Analysis 6-9
   d. Functional Requirements 10-11
   e. Non-Functional Requirements 12
   f. UML Diagrams 13-19
   g. Architectural Patterns 20
   h. Scientific/Academic Foundations 21
   i. Testing & Validation 21
   j. Recommendations for Future Development 22

Introduction:

Background & Problem Definition:

In many academic campuses, the management of shading assets and outdoor areas is still ad hoc and fragmented. Information is scattered across Excel spreadsheets, images, WhatsApp messages, and emails - with no single system of record. There is no common standard for map annotation, recording dimensions and coverage percentages, or logging installation/repair dates. The result is inconsistency across departments (Maintenance, Operations, Administration), difficulty in tracking site changes, and reliance on tacit knowledge held by a few individuals. Operationally, this produces delays and recurring issues:

- Tasks are handled late due to missing basics (which photo is the latest? Where exactly is the object located? What was the last recorded status?).
- Field teams arrive without a unified work plan.
- Prioritization by urgency or impact is difficult.

In addition, access control is unclear: documents and images are freely shared, updated without oversight, and sometimes stored locally without backup. Without a proper change history, auditing is hard, managerial reporting is limited, and forward planning for seasonal expansions or replacements is compromised.

Collectively, these gaps lead to wasted time and budget, reduced transparency for leadership and campus users, and a poorer on-site experience in areas that are insufficiently shaded.

Value Proposition and Justification:

- **Centralized, up-to-date mapping -** A single repository linking each Area to its base map and shading assets (location, dimensions, coverage %) reduces errors and provides a consistent operational picture for all stakeholders.
- **Streamlined maintenance workflows -** Creating, updating, and removing shading assets and area data becomes faster and standardized - capturing change logs and history - shortening turnaround times and improving prioritization.
- **Transparency and access control -** Clear separation between a System Administrator (admin) and read-only viewers (guest) prevents unauthorized changes and strengthens trust.
- **Reduced reliance on ad hoc files -** Uploading base maps to a managed storage location (e.g., an uploads directory) and storing the path on the Area record ensures consistency, backup, and straightforward version management.
- **Data-driven decisions and forward planning -** Harmonized data enables identification of problem areas, preventive planning, and improvements to the campus user experience (e.g., where shading is sufficient or lacking).

Key Terms and Definitions:

- **Area -** A campus mapping unit that serves as the organizational frame for managing and displaying the shading assets it contains.
- **Area Map -** The current image of an Area used as the visual base layer on which shading assets are marked and managed.
- **Shade -** An object representing a shading asset with a position and relative size on the Area Map.
- **Coverage Percentage -** A simple metric indicating how much of the relevant surface is shaded; used for status assessment and comparison across Areas.
- **System Users and Roles -** A basic role model distinguishing between a System Administrator and view-only users.
- **User Authentication -** A secure sign-in process that identifies the user and grants access according to role.
- **Role-Based Access Control (RBAC) -** A mechanism ensuring that only authorized roles can perform administrative actions, while others are limited to viewing.
- **Area Management Workflow -** A typical sequence: create a new Area, attach its base map, add Shades, and update them over time.
- **Visual Editing of Shades -** Intuitive adjustment of position and dimensions directly on the map to reflect on-site reality.
- **Search and Filter -** Tools for quickly locating an Area or Shade by name to improve efficiency and task prioritization.
- **Documentation and Currency -** Centralized data storage to reduce gaps and duplication and to ensure field teams see the most up-to-date version.
- **Media and Files -** Area images stored in an orderly, accessible manner for display and management—without relying on ad hoc, scattered files.

System Description:

Project Overview:

The Shade-System project is a web-based platform for managing and visualizing shaded areas across a campus.
The platform consolidates base-map images and shading points onto an interactive map, providing a clear, up-to-date visual status.
With the system, maintenance and planning teams can identify where additional shading is needed, improve the usability of outdoor spaces, and make informed decisions that enhance the on-campus user experience.

Feasibility Assessment:

The system is designed as an organizational platform that turns up-to-date shading data across the campus into actionable operational insights. It can be effectively adopted by educational institutions, facilities management organizations, and operational departments.

Contribution to Organizational Objectives:

- **Accessible shading information across campus -** The system presents clear, current data on shading locations so students and staff can quickly understand on-site conditions and use it in day-to-day activities and events.
- **Streamlined workflows and inter-department coordination -** By centralizing information in one place, the platform reduces repeated site visits and improves coordination among planning, maintenance, and administration units - enabling faster response and better service for students and staff.
- **Improved user experience and safety -** The platform supports informed planning of shaded areas for staying, learning, and activities - enhancing comfort and reducing sun exposure for the benefit of students and staff.
- **Data-driven decision-making -** A consolidated visual status by Area helps identify shading gaps and prioritize upgrades and maintenance according to real needs in the field.
- **Transparency and quality control -** Role-based access separates viewing from editing, preserving data consistency and accuracy while enabling change tracking - supporting service quality and clear communications with students and staff.
- **Innovative and sustainable image -** Demonstrates the use of advanced technology for managing outdoor spaces, projecting professionalism and innovation that strengthens the organization's brand. Creating comfortable shaded zones also reduces reliance on air-conditioning and supports responsible environmental practices.

Integration with Existing Organizational Systems:

- **Enterprise Identity Management -** The system will connect to the organization's user directory/database for authentication, ensuring access is granted only to organizational users in line with access policies.
- **Architectural/Planning Image Repository -** The platform will provide access to the organizational image library to select and set an up-to-date Area base map on which on-site Shades can be positioned.
- **Student Portal search and filter/Campus App -** The system can be embedded or linked to provide read-only views of shading status by Area—supporting transparency and guiding users throughout open campus spaces with a convenient, efficient interface.

Stakeholder Analysis:

A. Direct Stakeholders

1. Campus Leadership / Administrators

Campus leadership seeks a centralized tool for real-time, data-driven operational and budget decisions that presents the shading status of open campus areas at a glance. They need a system that quickly highlights gaps, helps prioritize investments and maintenance, measures project effectiveness, and reinforces an innovative and sustainable brand for students, faculty, and the public—all from a single, consistent, and controlled interface.

- Positive impact:
  - Data-informed investment prioritization **-** A consolidated visual map of shading status by Area enables rapid identification of hotspots and targeted allocation of budget and resources to actions that directly improve student and staff well-being.
  - Managerial transparency and coordination **-** Centralizing information creates a "shared language" across planning, maintenance, and administration units, reducing misunderstandings and shortening decision cycles.
  - Innovative, sustainable branding **-** A digital solution for managing outdoor campus spaces signals professionalism and environmental responsibility, aiding student recruitment and retention.
- Potential downsides:
  - Onboarding resources and timelines **-** Rollout requires defining Areas, uploading images, and formalizing procedures. Advance planning for resourcing, roles, and dependencies is needed to minimize disruption to parallel projects.
  - Change resistance **-** Moving to new workflows can encounter preference for legacy methods, skepticism, and reluctance to adopt new procedures.
  - Data maintenance burden **-** Decision quality depends on keeping maps and data up to date; delays or neglect lead to partial views and divergence from on-site reality.

2. Students and Faculty

Students and faculty want a tool that simplifies daily navigation on campus—planning routes, breaks, and seating in comfortably shaded locations—supporting well-being, focus, and heat safety throughout the day. They need an accessible, real-time view of shading levels at any point on campus to quickly choose where to stay or how to walk based on their schedule

Positive impact:

- Efficient day planning **-** Rapid selection of "cooler" walking paths, meeting points, and shaded seating by time and location saves time and reduces fatigue and heat exposure.

- Improved comfort and focus - Staying in shaded areas reduces glare and extreme heat, supports comfort, and enables studying or preparing lectures in open spaces between classes.
- **Transparency and accessibility:** An updated mobile map provides control and a simple experience, helping users understand where it's most comfortable right now.

- Potential downsides:
  - Dependence on data freshness - If information is outdated, users may experience a gap between what they see and on-site conditions.
  - Expectations for immediate fixes - Visibility of unshaded areas can raise expectations for rapid remediation, creating temporary frustration until work is completed.
  - Technology distraction - Overreliance on the app (frequent checks, monitoring coverage percentages) may divert attention from day-to-day on-site awareness.

3. Campus Maintenance & Operations Department

Maintenance teams need a centralized platform showing real-time shading status across campus Areas to rapidly identify gaps, prioritize field tasks, plan efficient work cycles, and measure "before/after" outcomes.

- Positive impact:
  - Accurate planning and prioritization - Shading status by Area reduces unnecessary inspection rounds and supports focused daily/weekly work plans.
  - Control and performance measurement - Change logs provide a transparent basis for evaluating work, shortening approval cycles, and presenting progress to management.
  - Cross-department alignment - A shared Area map creates a common reference that reduces misunderstandings among maintenance/operations sub-teams (gardening, electrical, installers, etc.) and improves resource and scheduling coordination.

- Potential downsides:
  - Initial data-entry overhead - The initial consolidation (defining Areas, uploading images, placing Shades) can temporarily slow routine operations.
  - Reliance on data accuracy - Inaccurate or late updates to on-site shading status may lead to misguided decisions.
  - Sensitivity to transparency metrics - High visibility of statuses and timelines may be perceived as excessive oversight, causing discomfort within teams.

B. Indirect Stakeholders

1. Campus Planning & Development Division

The organization's planning and development arm seeks to pinpoint where infrastructure or design changes are needed and to embed shading standards into future development plans—grounded in clear, consolidated data.

- Positive impact:
  - Sharper capital prioritization - Frequent identification of under-shaded areas helps focus development budgets on high-impact locations.
  - Up-to-date design specifications - Using current maps and field notes to define clear requirements for shading deployment in new projects.
  - Equity across campus zones - The system highlights disparities between faculties/zones, supporting fair, consistent principles for allocating shading projects.
- Potential downsides:
  - Reordering existing plans - New data may challenge previously set priorities, creating friction with stakeholders who counted on established budgets.
  - Pressure for frequent plan updates - High transparency about gaps can drive calls for more frequent design revisions than typical development cycles allow.
  - Short-term planning bias - Relying on a point-in-time map view may push "quick local fixes" at the expense of deeper, long-term solutions, prompting repeated redesigns later.

2. Marketing, Communications, and Student Recruitment

These teams aim to present a smart, sustainable, student-centric campus and to craft narratives that demonstrate tangible improvements in the experience of open spaces.

- Positive impact:
  - Innovative environmental positioning - Showcasing proactive shading management supports a sustainability-and-care narrative, differentiating the institution and resonating with applicants and partners.
  - Synergy with events and campaigns - Coordinating shoot times and event locations in comfortably shaded areas improves on-site experience and visual outputs.
  - Higher engagement via real-time, useful content - Turning shading data into brief updates produces immediately valuable content that drives reactions, shares, and traffic—positioning the campus as a day-to-day helpful brand.

- Potential downsides:
  - Exposure of under-shaded zones **-** Transparency may highlight areas not yet addressed, requiring careful timing and examples to avoid reinforcing a “gap” narrative.
  - Rising expectations - After publicizing improvements, audiences may expect rapid, campus-wide upgrades. If implementation is slower or scoped, this can be perceived as inconsistency.
  - Message fatigue **-** Frequent posts about shading updates and locations can feel repetitive, reduce interest and engagement, and dilute the impact of high-priority messages when needed.

3. External Vendors and Contractors

Vendors and contractors seek clarity and focus—knowing where and when solutions are needed to price accurately, schedule crews, and minimize time on site. The system serves as a shared reference with the client, speeding the path from quotation to execution.

- Positive impact:
  - Targeted, measurable orders - Requirements grounded in current conditions reduce “gray areas” in specs and quantities, enabling accurate bids and fewer disputes about what was ordered versus delivered.
  - Clear coordination and **timelines -** Predefined work windows derived from the needs map cut unnecessary on-site waiting, improving crew/equipment utilization and stabilizing overall costs.
  - Defined quality metrics **-** Point-specific before/after documentation anchors acceptance checks, reducing subjective interpretation and speeding final account settlement.

- Potential downsides:
  - Dependence on access permits and site coordination **-** Even with a clear map and priorities, delays in site access, temporary blockages, or overlaps with other works can disrupt execution and create unplanned downtime.
  - Seasonal demand volatility **-** Heat waves and summer months drive peak orders and tight schedules, while quieter seasons drop sharply - making steady staffing and resource allocation harder.
  - Safety and operating-hours limits - To avoid disrupting teaching and campus life, work may be constrained to off-peak or limited hours - shrinking effective work windows and lengthening project timelines.

Functional Requirements:

1. **User Registration & Authentication**
   Behavior - Allow campus users to register by validating basic identity fields: name, password, phone, and email.
   System responses - Display an error if a required/invalid field is detected or if the user already exists.
   Constraints - No duplicate registration with the same username or email.

2. **Area List View**
   Behavior - Load all Areas from the server and display a record per Area.
   System responses - Show “No results” when there are no matches or no available Areas.
   Constraints - Dependent on server availability.

3. **Area Search**
   Behavior - Text search by Area name within the list.
   System responses - Live filtering while typing; display “No Areas found” if there are no matches.
   Constraints - Basic text-based search.

4. **Create New Area**
   Behavior - Add a new Area with fields: name, description, and a base-map image.
   System responses - Validate required fields; show a success message and refresh the list, or show a targeted error.
   Constraints - Area name must be unique; the image must be valid and under a defined maximum size.

5. **Upload/Replace Area Image + Preview**
   Behavior - Select an image file to serve as the Area Map and preview it before saving.
   System responses - Show an error for invalid files.
   Constraints - Supported formats only; maximum file size enforced. Validation is performed **server-side** (MIME type and extension whitelist).

6. **Delete Areas**
   Behavior - Select and delete a single Area or multiple Areas.
   System responses - Show an explicit confirmation prompt and a success/failure message.
   Constraints - Irreversible action; require a second confirmation for bulk deletion.

7. **Navigate to Area Editing**
   Behavior - Open a dedicated Edit screen when clicking an Area name.
   System responses - Load the Area’s data into the Edit view.
   Constraints - The Area must exist in the database; otherwise show an error.

8. **Consolidated Area Edit Mode**
   Behavior - Enter an Edit screen that allows updating the Area name/description and modifying multiple Shade points in a single session.
   System responses - Prominent “Edit mode” indicator; success/failure message on save.
   Constraints - Area name must remain unique; deletions require confirmation; no version history is maintained.

9. **Add Shade Points on the Image**
   Behavior - Clicking “Add Shade” opens a form and lets the user place a Shade point precisely on the map.
   System responses - Icon appears immediately; open the Shade properties form; allow cancel/exit without saving.
   Constraints - Allowed for the Admin profile only, and only when the Area exists and its map has successfully loaded. A Shade point may be placed exclusively within the map boundaries. Supported formats only; enforce a maximum file size

10. **Shade Point Properties**
    Behavior - Set a Shade description and a coverage percentage between 0–100; allow dragging the point on the map or positioning it by coordinates.
    System responses - Validate numeric ranges and coordinate bounds; reflect changes instantly in both the icon and the form.
    Constraints - A valid Area image is required; the number of points per Area is capped by a system rule.

11. **Delete Shade Points**
    Behavior - Remove an individual Shade point from the Edit screen.
    System responses - Show a confirmation dialog before deletion and a status message afterward.
    Constraints - Irreversible action; explicit confirmation required.

12. **Visual Coverage Indicator**
    Behavior - Display a “filling” icon per Shade reflecting its coverage percentage, with a clear legend.
    System responses - Update the indicator in real time when coverage or position changes.
    Constraints - Must comply with accessibility guidelines.

13. **Save Area-Wide Changes**
    Behavior - Persist all Area and Shade updates to the database and show a save indicator.
    System responses - “Saved successfully” message or a network error with a retry option.
    Constraints - No historical reports or period-over-period comparisons.

14. **Full Mobile & Desktop Support**
    Behavior - Responsive layout; full usability on touch devices and desktops.
    System responses - Automatic UI adjustments based on screen size while preserving workflow.
    Constraints - Modern browsers only.

15. **Server API for Areas/Shades**
    Behavior - Expose endpoints to load, create, update, and delete Areas and Shade points.
    System responses - Consistent HTTP status codes and JSON payloads with clear error messages.
    Constraints - Dependent on database/file storage; no analytics or reporting included.

Non-Functional Requirements:

1. **Consistent Response Times**
   The system maintains short, stable response times under light-to-moderate load, ensuring routine tasks run without noticeable delays and the user experience remains smooth.

2. **Secure Database Access**
   Database operations use parameterized inputs, separating query logic from supplied values. This mitigates injection risks and preserves data integrity and consistency over time.

3. **Clean File Management**
   Uploaded files receive unique identifiers to prevent overwrites and duplicate names. Unused files are not left "orphaned" in storage, keeping the repository orderly and space-efficient.

4. **Clear, Consistent API Responses**
   All endpoints return a uniform JSON structure with explicit status codes and messages. This consistency enables predictable error/success handling and simplifies integration and maintenance.

5. **Client-Side Privacy**
   Only the minimum data required for the user experience (e.g., display name) is stored on the client. Sensitive details are not retained client-side, reducing exposure in case of unauthorized access to the device.

6. **Efficient Browser Resource Usage**
   The application avoids holding unnecessary transient resources (views/files) in memory, preventing buildup that can cause slowdowns during continuous use.

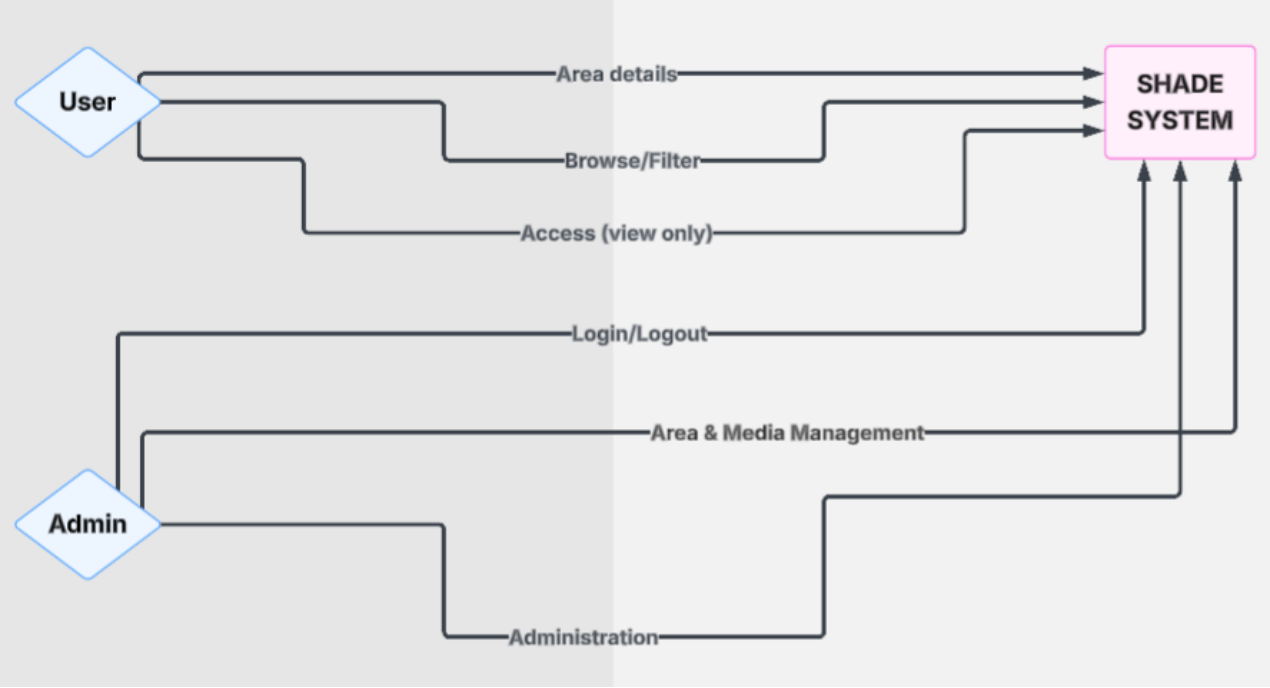
7. **Screen Adaptation and Directionality**
   The UI adapts to a wide range of sizes and orientations and supports Hebrew (RTL). Layouts, readability, inputs, and components behave correctly across relevant screen scenarios.

8. **Controlled Cross-Domain Compatibility**
   The system supports reliable interaction between the front end and the API even when served from different origins, ensuring proper loading and network requests in line with browser policies (e.g., CORS).

9. **Maintainability and Extensibility**
   The system follows a layered architecture (client, server, database) with modular components. This structure eases fixes, testing, and the addition of new capabilities without impacting the entire system.

UML Diagrams:

1. Context diagram:



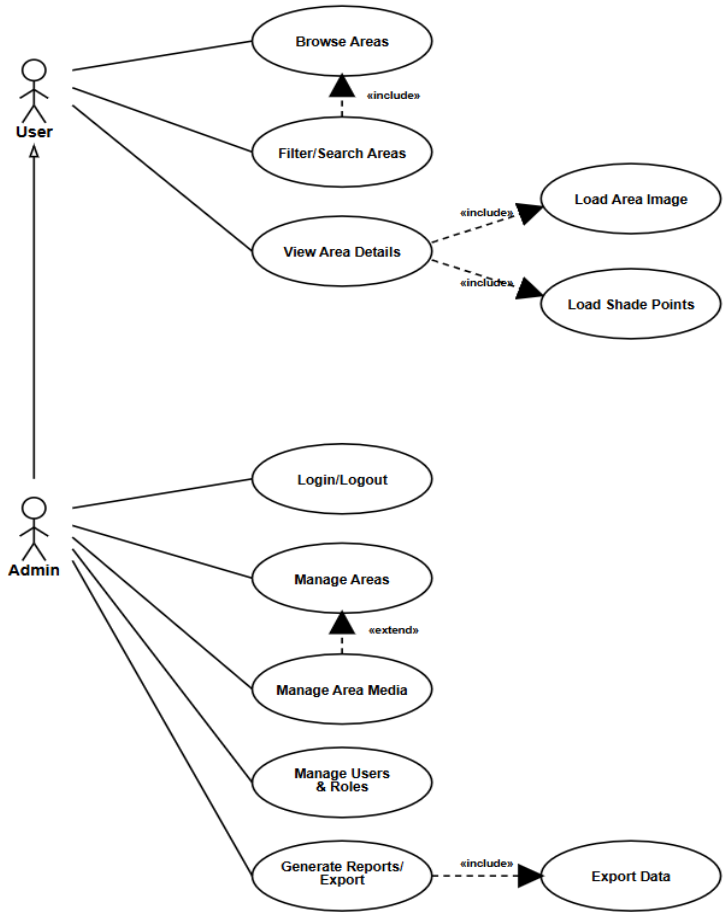
Explanation

The diagram outlines the system boundary of the **Shade System** and its two external actors: **User** and **Admin**.
The connectors indicate each actor's interactions with the system:

- The User has read-only access - browsing Areas, viewing details, and applying filters.
- The Admin manages Areas and media assets and performs general operational actions.

In essence, the diagram emphasizes that the User is a consumer of information without edit privileges, whereas the Admin is responsible for creating, updating, and administering content and configuration across the system.

2. Use case diagram:



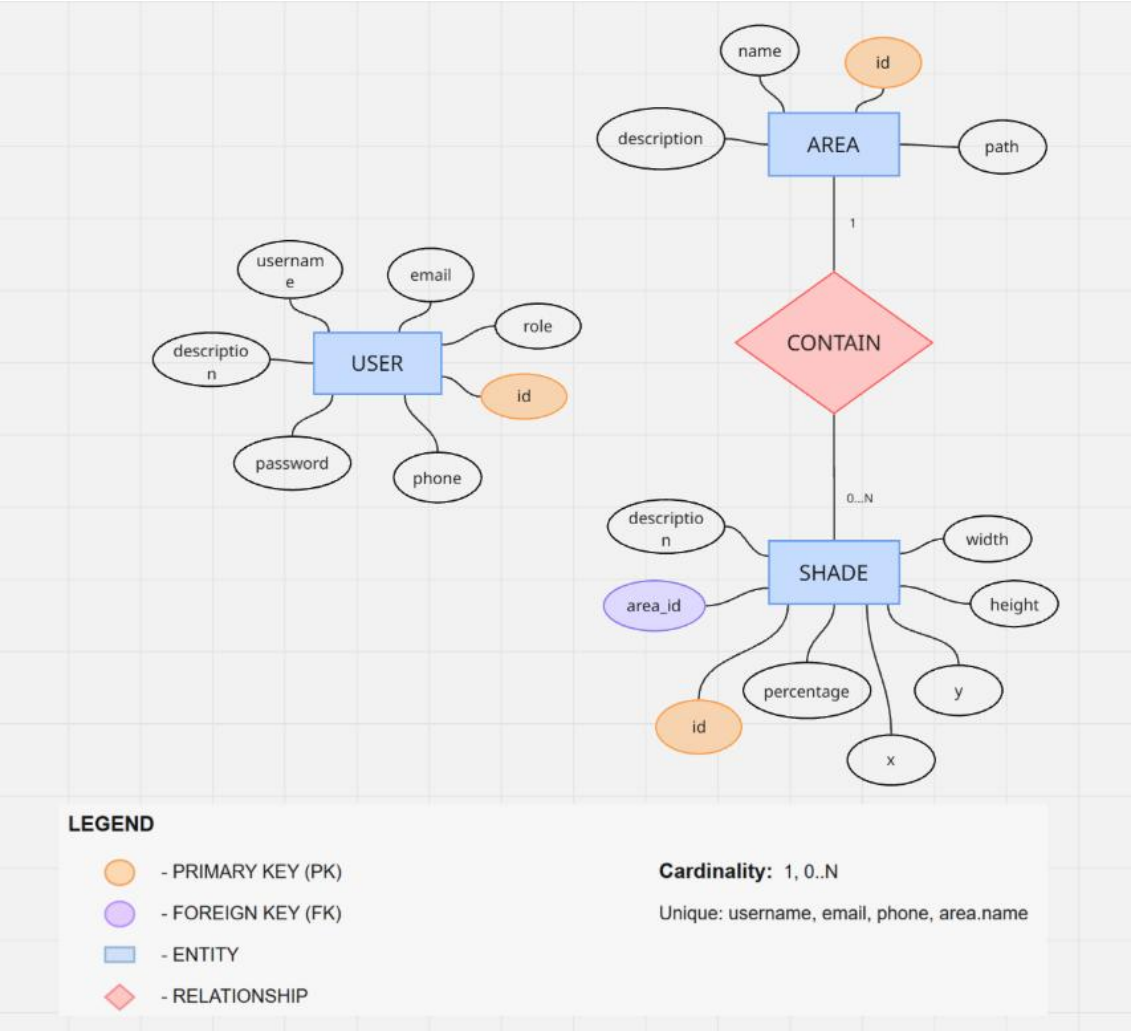
Explanation

The diagram maps use cases by actor.

- **End User (User):** Navigates between Areas, performs search and filter, and views Area details—including loading the Area image and the associated Shade points.
- **System Administrator (Admin):** Manages sign-in/sign-out, oversees Areas (including media handling), and administers platform users.

This clearly distinguishes read-only viewing for end users from administrative and maintenance actions performed by the Admin.

3. ERD:



Data Model Explanation

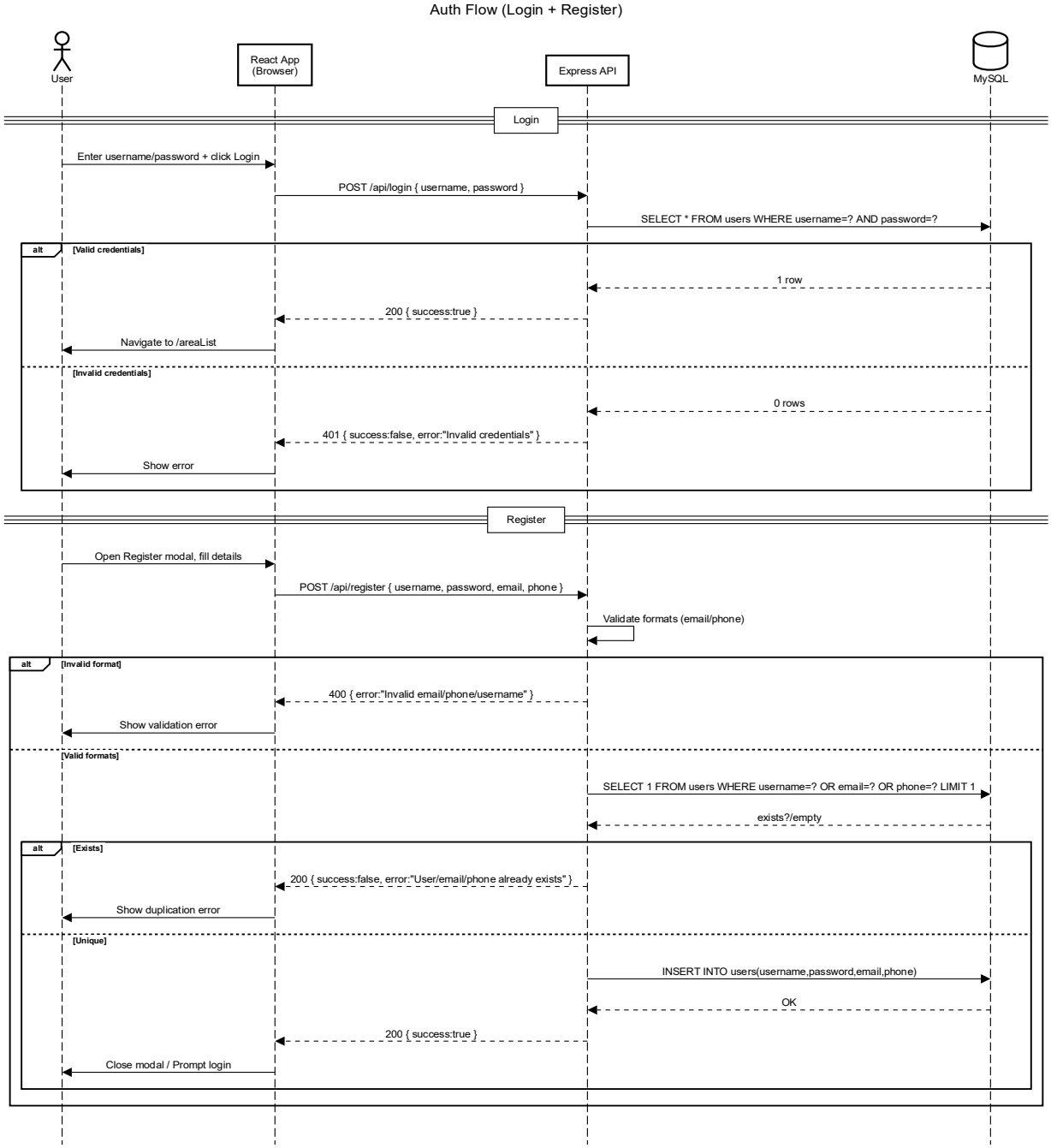
The diagram presents three entities: **USER**, **AREA**, and **SHADE**.

- USER holds identifying fields and profile attributes.
- AREA stores all Area details (including a reference to the Area's base-map image).
- SHADE contains shading properties and a foreign-key link to its parent Area.

The **CONTAIN** relationship specifies that a single AREA can include multiple SHADE records (i.e., each Shade belongs to exactly one Area).

The legend also indicates primary keys, foreign keys, and uniqueness constraints in the database—ensuring, for example, unique usernames and unique Area name scopes.

4. Sequence diagram

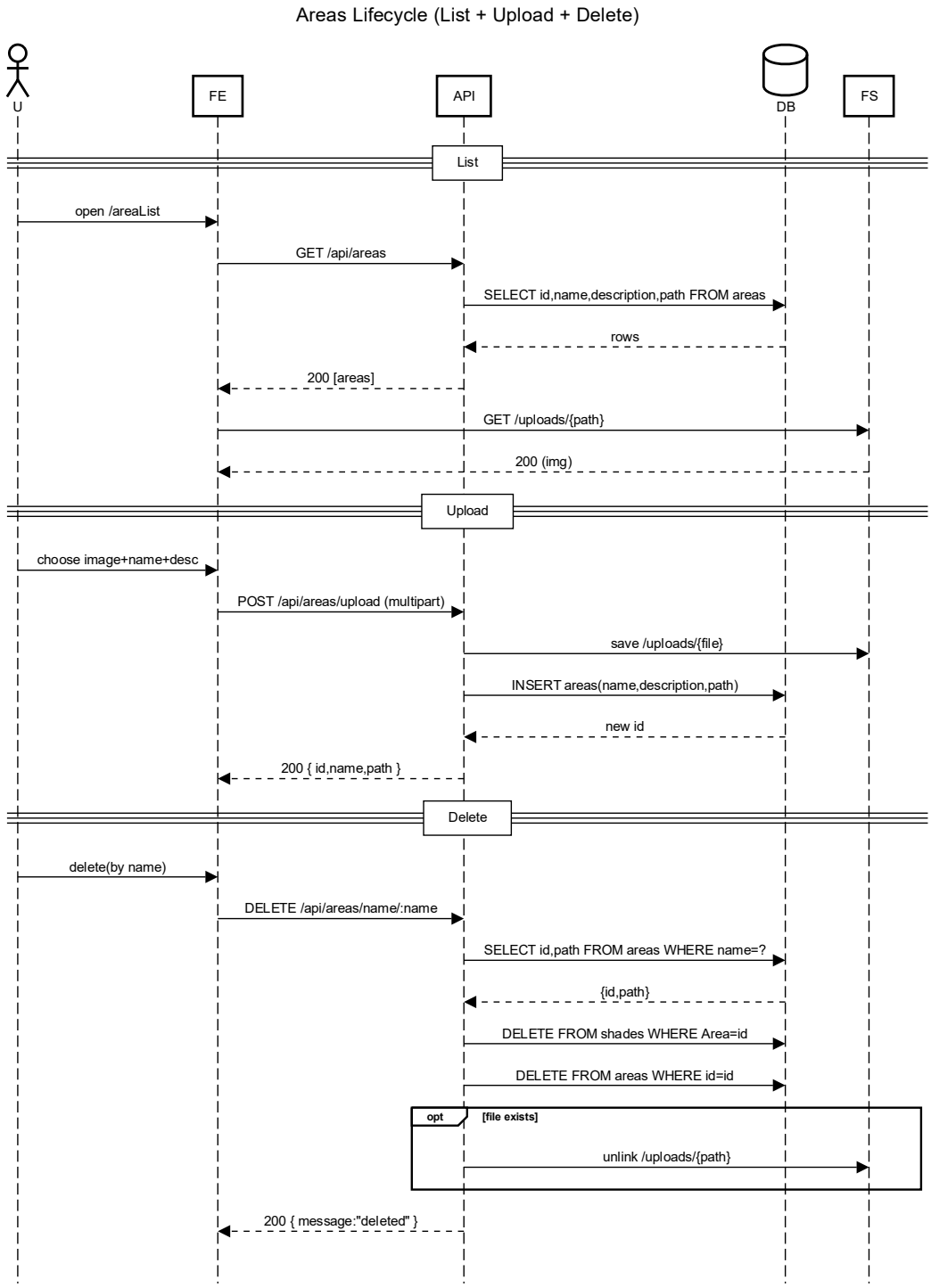


Explanation

The diagram illustrates the authentication flow across the User, the browser client, the Express server, and the database.

- **Sign-in:** The system validates the submitted credentials and returns a success or error response.
- **Sign-up:** The system performs input validation and duplicate checks; if all checks pass, it creates a new user with a “guest” role.

Directional arrows indicate the chronological sequence of requests and responses exchanged between components over time.

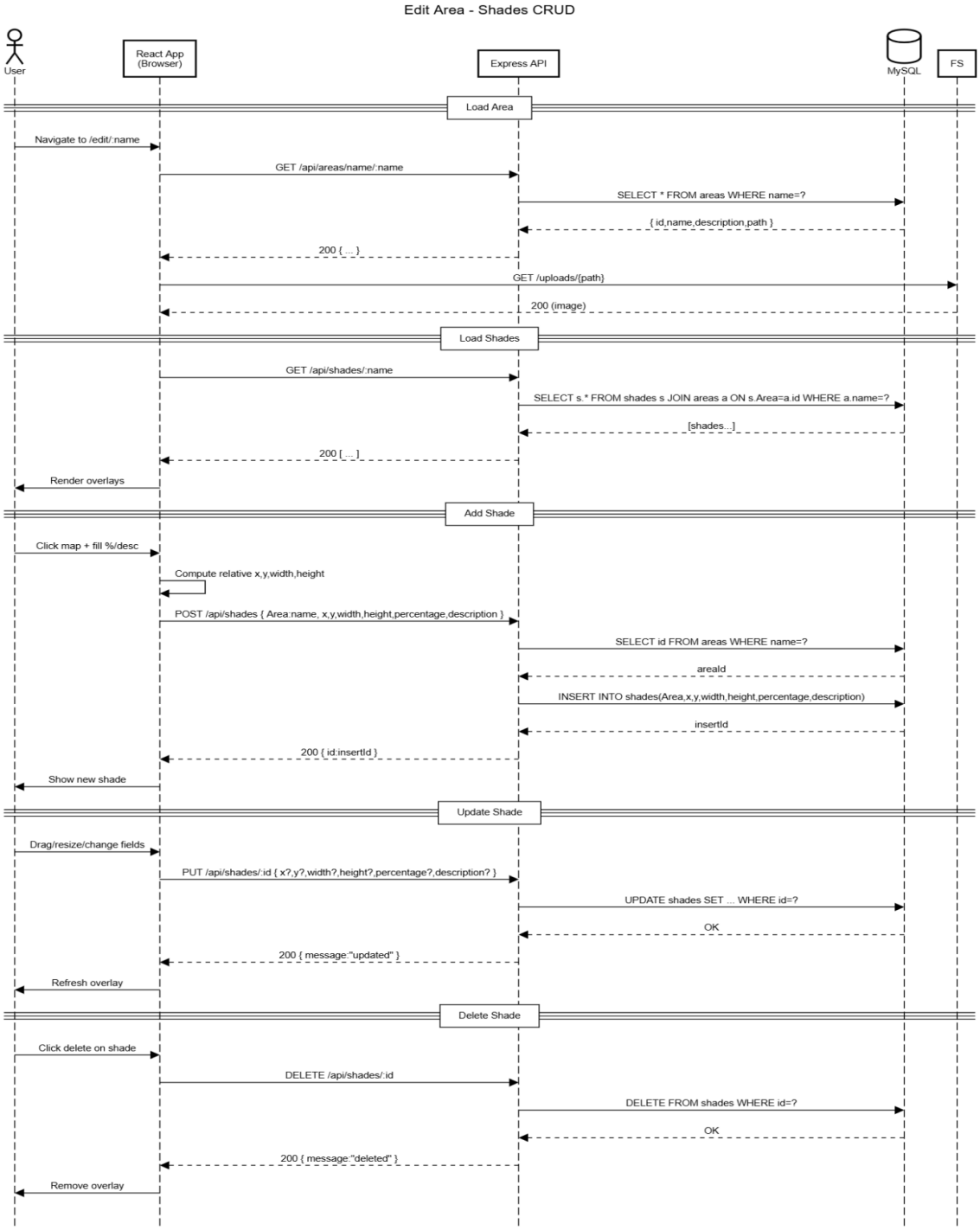


Explanation

The diagram depicts the lifecycle of Areas in the system (list, create, and delete) and the coordination among the client, the API server, the database, and the file system.

- **List:** The client requests Area data from the server; the server returns the records, and the UI displays the Area names.
- **Create:** The client submits the Area name, description, and base-map image. The image is stored in the file system, the Area record is persisted in the database, and the UI shows a preview of the uploaded image.
- **Delete:** Deleting an Area also removes its associated Shades and dependencies.

Directional arrows illustrate the chronological sequence of requests and responses exchanged between components over time.

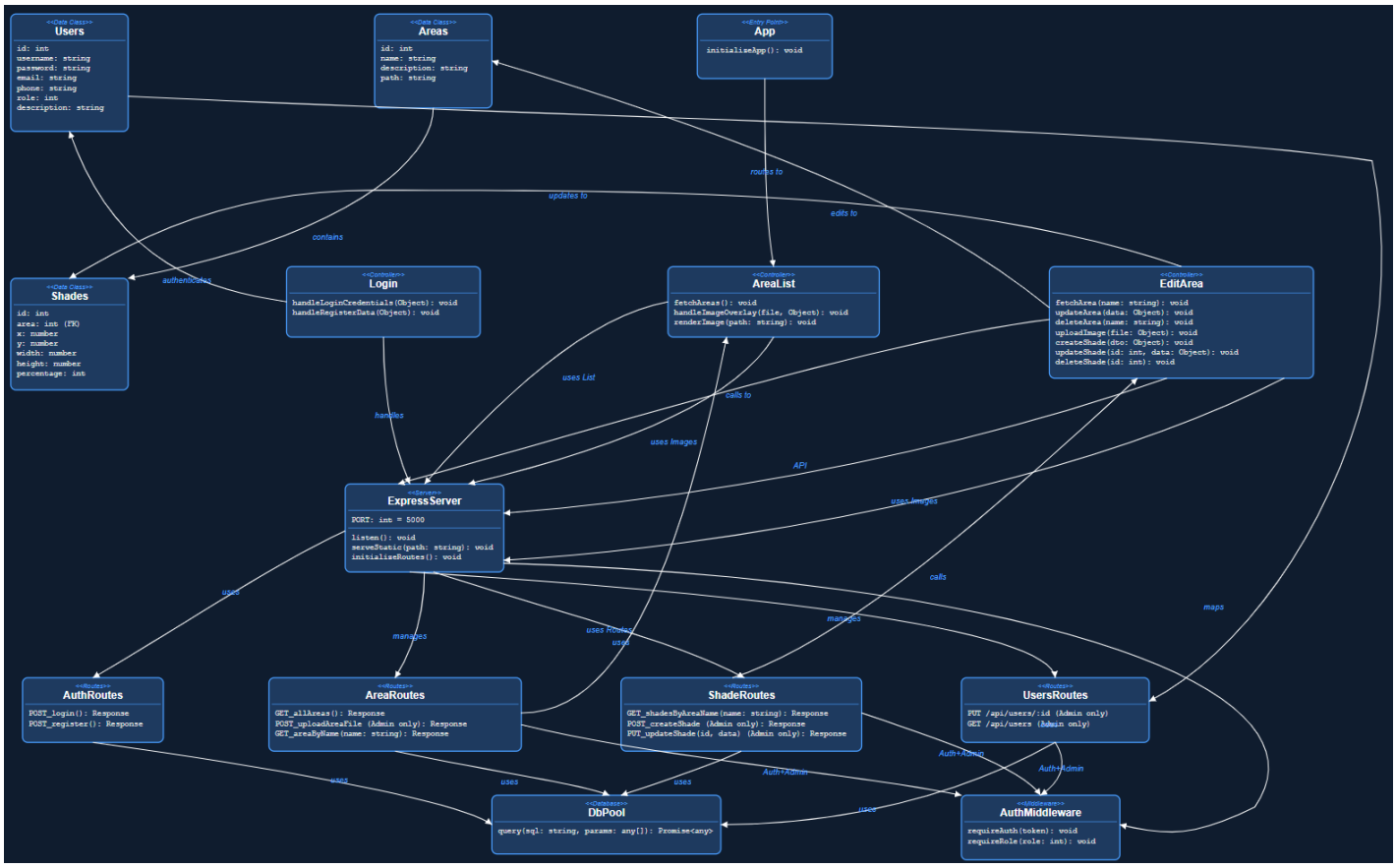


Explanation

The diagram shows the Area Edit workspace for managing shading. The system loads the selected Area and its base-map image, retrieves the associated Shades, and renders their layers on the map.

- **Create (Add Shade) -** Click on the map and enter the Shade details; the system calculates and stores the Shade's position and dimensions relative to the image.
- **Update (Edit Shade) -** Modify the relevant fields; the system saves the changes and refreshes the Shade layer in the view.
- **Delete (Remove Shade) -** Delete the selected Shade; it is removed from the display and its record is deleted from the database.

5. Class diagram:



Explanation

The diagram presents the end-to-end system architecture and the connections between the Frontend, Backend, and Database layers.
On the client side, the React components include the main application (App), the sign-in screen (Login) for authentication, the area list (AreaList) for displaying Area names, and the area editor (EditArea), which performs create/read/update/delete operations on Shades.
These components communicate with the Express server through a dedicated routing layer: AuthRoutes for authentication, AreaRoutes for Areas, ShadeRoutes for Shades, and UsersRoutes for user management (admin only).
On the data layer, the DBPool executes queries against the database tables. Together, this yields a unified flow: the UI initiates actions, the API enforces business logic and access control, and the database persists the information.

Architectural Patterns:

1. Client–Server Architecture

- General explanation:
  This architecture separates the client—which renders the user interface and issues requests—from the server, which centralizes business logic, security, and data access. The client sends service requests to the server; the server processes them, accesses the database or file storage as needed, and returns a response—typically in JSON. This separation enables independent development, deployment, and scaling of each layer, consistent security policy enforcement in one place, and standardized coordination through a stable API.

- Implementation in the Shade System:
  In the Shade System, the client layer is a browser-based React application responsible for presentation, navigation, and forms (sign-in, Area list, and Area editing). The client issues HTTP requests to an Express-based API and receives JSON responses. The server layer consolidates business logic, user authentication and authorization, file handling, and database access. Communication between layers occurs via clear, well-defined endpoints, keeping the UI focused on presentation while the server enforces rules, security, and long-term data persistence. This separation allows each side to be developed, tested, and deployed independently, while maintaining a stable API that serves current and future screens and features.

2. MVC (Model–View–Controller) Pattern

- General explanation:
  MVC is an architectural pattern that divides the system into three complementary components:
  Model – represents data, business rules, and database access.
  View – the presentation layer responsible for the user interface and visual rendering.
  Controller – the intermediary that receives input from the user/client, interprets the request, invokes Model logic, and returns a result to the View.
  Separating data, logic, and presentation reduces coupling, simplifies maintenance and testing, and enables system growth without breaking other layers.

- Implementation in the Shade System:
  On the server side, the Shade System applies an API-oriented variation of MVC:
  Controller implementations define HTTP routes, validate input and permissions, and dispatch requests to the data layer.
  Model components implement database access and uphold relational consistency.
  View is expressed as structured JSON responses rather than HTML pages—the server returns a data representation that the client renders.
  Cross-cutting middleware (e.g., authorization, file handling, error handling) complements the pattern by centralizing concerns that span multiple routes.
  The result is a clean division: presentation on the client, control and business logic in Controllers, and durable data modeling in Models.

Scientific/Academic Foundations:

The project is grounded in engineering principles that ensure accuracy, consistency, and data durability.
Shade points are stored as coordinates relative to the Area's base-map image (percent of width/height), enabling precise, responsive rendering across all resolutions and devices.
Each point maintains a coverage percentage (0–100) as a simple metric for comparison and prioritization.
Data integrity is enforced through parameterized queries, primary/foreign keys, and uniqueness constraints (e.g., Area/User names).
Role-Based Access Control (RBAC) applies the principle of least privilege, separating read-only users from administrators.
File handling employs unique identifiers, server-side validation (type/extension and size), and cleanup of replaced or deleted files to prevent orphaned artifacts.

Testing & Validation:

Quality was validated across multiple layers.
Server: API integration tests verify core routes (Areas/Shades/Auth) under success, edge, and error scenarios against a seeded test database, asserting consistent status codes and JSON structures.
Client: Smoke tests cover the critical user flows—sign-in/sign-out, Area listing and search, and add/update/delete of Shade points—along with responsive checks for common mobile and desktop breakpoints.
Security: Role-based authorization is enforced for all state-changing actions; all database access uses parameterized queries; and file uploads undergo server-side validation (type/extension and size).
Performance: Targets short response times under light-to-moderate load and verifies image uploads within supported size limits.
Operations: Client-side session data is fully cleared on logout, and server logs are used to track anomalies.

Recommendations for Future Development:

- **Integration with Moodle**
  Integrate the Shade System via the institution's Single Sign-On or a native Moodle integration (e.g., block/plugin widget). Moodle will pass the authenticated identity and role to the Shade System, eliminating the need for an additional login. Within Moodle, the widget will display a small map/image of the selected Area and a short list of recommended shaded spots, with a link to a full view. This approach centralizes authentication/authorization, increases adoption (users already live in Moodle), and gives management broader reach, a unified data layer, and interaction from a single institutional platform.

- **Heatmaps and Simple Charts**
  Aggregate existing data into a dedicated table and compute daily/weekly statistics. On the client side, present a "heatmap" that highlights problematic Areas and concise charts showing trends throughout the day. Offloading calculations to batch jobs (rather than on every page load) keeps the UI fast and lightweight. For users, raw data becomes intuitive insight; for management, this provides an immediate operational picture to guide decisions—where to invest, what has improved, and which Area is most urgent.

- **Open Service Tickets on the Map**
  From the Area screen, clicking a point on the map opens a "New Ticket" form (short description, phone photo upload, urgency, and Area/Shade tag). The ticket is recorded in the database and shown as a graphical layer on the map, with filters by status and time. This can later expand to integrate with the institution's existing ticketing system. Keeping every request in spatial context reduces location errors and standardizes the workflow. Users get a simple, quick process; management gains transparent tracking of workload and recurring requests.

- **Smart Notifications**
  Define lightweight rules that trigger alerts: significant Area changes (add/remove), overdue tickets, or coverage targets falling below a threshold. A server-side scheduler runs at short intervals, evaluates conditions, and sends email/mobile notifications by role. This avoids "notification noise": only important events reach the right people. Operationally, there is less need to chase manual updates, and both users and management experience continuous control—critical issues are raised in time, enabling real-time action rather than after-the-fact reports.